

Validity Analysis: CRISISLTLSUM

Target context: Mexico City and Central America Bilingual Crisis Journalism

Overall risk: **HIGH**

Deployment Context

Use case: Summarization of quick messages (e.g. tweets) for event reporting in the crisis domain for news writing in American English and Mexican Spanish mostly

Target population: Journalists and media workers who monitor social media to cover breaking news from Mexico City. They would speak Mexican Spanish natively (possible neutral/standard regional variety), and English fluently. They would be expected to use the model in English and Spanish to cover news of disasters happening not only in Mexico, but also south of the United States and also Central America.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	1	Serious concern	high
Input Content 	1	Serious concern	high
Input Form 	3	Moderate gaps	medium
Output Ontology	2	Significant gaps	high
Output Content 	1	Serious concern	high
Output Form 	1	Serious concern	high
Average	1.5		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

CrisisLTLSum is a fundamentally English-language, US/Australian-anchored crisis-tweet benchmark whose four-domain ontology (wildfire/local fire/traffic/storm) [Q1, Q122], English-keyword-filtered content [Q126], English NER + OpenStreetMap pipeline [Q35, Q36], Anglophone MTurk annotators [Q61], English-anchored coherence rubric [Q153], and English-only ROUGE evaluation [Q99, Q100] together create severe validity gaps for the target deployment of bilingual crisis journalism in Mexico City and Central America. The structurally important crisis types for the deployment area — earthquakes, volcanic

activity (Popocatépetl), tropical-system flooding, civil unrest, and CDMX transportation infrastructure failures — are entirely absent from the benchmark [WEB-7, WEB-20, WEB-23]. The Spanish-language output requirement (one of two deployment languages) receives zero coverage from the benchmark's evaluation framework. Of the six dimensions, five are scored 1–2 (high concern) and only Input Form scores 3 due to modality match. The authors themselves acknowledge non-portability to other languages and regions [Q119].

ID	Evidence
Q1	"CrisisLTLSum contains 1,000 crisis event timelines across four domains: wildfires, local fires, traffic, and storms." (p.1)
Q35	"Since location mentions are crucial in extracting local events and existing models have low performance detecting them from noisy tweets text, we further developed a BERT-based NER model tuned on Twitter data to detect location mentions." (p.4)
Q36	"Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses." (p.4)
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q99	"The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels." (p.8)
Q100	"We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores." (p.8)
Q119	"Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies." (p.10)
Q122	"We build CrisisLTLSum by limiting the search to four crisis domains: wildfire, local fire, traffic, and storm." (p.12)
Q126	"We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past." (p.13)
Q153	"For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-20	CENAPRED semáforo phase vocabulary is essential output-ontology content for volcanic crises (https://www.cenapred.unam.mx/reportesVolcanesMX/)
WEB-23	#Verificado19S citizen-verification genre would be judged differently by Mexican journalists than US MTurk workers (https://arxiv.org/abs/2007.05848)

Practical Guidance

What This Benchmark Measures

CrisisLTLSum measures, for English-language US/Australian tweets in four crisis domains (wildfire, local fire, traffic, storm), how well systems can (a) extract relevant non-redundant updates from a noisy tweet stream given a seed tweet [Q5, Q83] and (b) produce abstractive summaries scored against multi-reference human summaries [Q98, Q99]. Within that English/US scope, it provides defensible signal: 90%+ inter-annotator agreement [Q77] and a 73.51% vs. 88.98% model-vs-human gap [Q96] indicate the tasks are non-trivial and well-instrumented. For the target deployment, the benchmark's English-output evaluation framework can serve as a starting point for the English half of the bilingual requirement.

Construct Depth

Construct depth is shallow for the target deployment. The benchmark probes timeline extraction and summarization only on English tweets in four event types with no Spanish content, no journalist annotators, no editorial-style axes, no semáforo or SASMEX vocabulary, and no register-appropriateness metric. Even where the benchmark provides defensible signal (English summarization quality on fire/traffic/storm events), it does not address whether systems handle Mexican Spanish, voseo, code-switching, alcaldía/colonia toponymy, CENAPRED/SSN attribution, the #Verificado19S verification-tweet genre, or AP Español register conformance. The Coherence rubric [Q153] is structurally inapplicable to Spanish output. Strongest dimension is Input Form (modality match); weakest are Input Ontology, Input Content, Output Content, and Output Form — all of which would need substantial supplementation.

What Else You Need

A complete assessment requires: (1) construction of a Spanish-language crisis tweet corpus anchored to Mexican/Central American events covering earthquakes, volcanic activity, tropical flooding, and civil unrest (no such corpus currently exists per [WEB-3, WEB-4]); (2) re-annotation by Mexican journalist annotators using an extended relevance/informativeness rubric that accounts for verification posts [WEB-23] and community solidarity posts; (3) a Spanish coherence rubric and AP Español-aligned register/attribution axes to replace [Q153]; (4) a bilingual summarization evaluation protocol using BERTScore-multilingual and LLM-as-judge alongside ROUGE [WEB-17, WEB-18]; (5) Spanish gazetteer integration for alcaldía/colonia/municipio resolution to replace the OSM-English pipeline [Q36]; and (6) explicit handling of SASMEX/CENAPRED hashtag and semáforo vocabulary as seed-tweet types [WEB-7, WEB-20].

ID	Evidence
Q5	"The timeline extraction task is formalized as: given a seed tweet as the initial mention of a crisis event, extract relevant tweets with updates on the same event from the incoming noisy tweet stream." (p.1)
Q36	"Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses." (p.4)
Q77	"The average timeline-level agreement between those two workers is 90.06%." (p.6)
Q83	"Given an initial seed tweet t_0 , and following chronologically sorted tweets t_i , $i = 1, \dots, n$, the goal of the timeline extraction task is to determine whether tweet t_i is part of the timeline, i.e., related to the same event and adding new information compared to all prior tweets $[t_0, \dots, t_{i-1}]$." (p.6)
Q96	"The best timeline extraction model achieves 73.51 average timeline accuracy, whereas the human-level performance is 88.98." (p.8)
Q98	"We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004)." (p.8)
Q99	"The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels." (p.8)
Q153	"For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)
WEB-3	HumAID is English-only despite including a 2017 Mexico earthquake subset — confirming the gap in Spanish-language Mexican crisis tweet content (https://crisisnlp.qcri.org/humaid_dataset.html)
WEB-4	Curiel et al. (PMC 2019) is the only known Mexican-Spanish crisis NER/geolocation resource and is not a timeline/summarization dataset (https://pmc.ncbi.nlm.nih.gov/articles/PMC6484392/)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-17	Standard ROUGE variants correlate weakly with human judgments for Spanish summarization; BERTScore-precision with multilingual models is recommended (https://arxiv.org/pdf/2503.17039)
WEB-18	LLM-as-judge approaches show strong correlation with human judgments in multilingual summarization evaluation (https://aclanthology.org/2024.emnlp-main.1085.pdf)
WEB-20	CENAPRED semáforo phase vocabulary is essential output-ontology content for volcanic crises (https://www.cenapred.unam.mx/reportesVolcanesMX/)
WEB-23	#Verificado19S citizen-verification genre would be judged differently by Mexican journalists than US MTurk workers (https://arxiv.org/abs/2007.05848)

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: CRISISLTLSUM | Context: Mexico City and Central America Bilingual Crisis Journalism

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's input ontology is restricted to four crisis domains — wildfire, local fire, traffic, and storm [Q1, Q53, Q122] — explicitly chosen because they are the most frequent in the selected US and Australian states [Q124]. For the target deployment, the elicitation and regional research identify earthquakes, volcanic activity, tropical-system flooding, civil unrest, and CDMX-specific transportation infrastructure failures as structurally important crisis types. Earthquakes are central to CDMX coverage given the 1985, 2017, and 2022 events, and Popocatepetl produces near-continuous low-level activity that yields a constant tweet stream rather than discrete spike events [WEB-7, WEB-20, WEB-26]. Even nominally overlapping categories (storms) carry different regional profiles: Mexican/Central American tropical systems use distinct alert vocabularies (semáforo phases, SMN/CONRED naming) that are not represented [WEB-19]. The 'local' definition (building/street/county, short-lived) [Q12] also conflicts with persistent volcanic phenomena and multi-day tropical systems. The taxonomy is therefore both under-representative (missing required categories) and structurally mismatched on the categories it does include.

ID	Evidence
Q1	"CrisisLTLSum contains 1,000 crisis event timelines across four domains: wildfires, local fires, traffic, and storms." (p.1)
Q12	"Moreover, we focus on the extraction of local crisis events. The term 'local' indicates that an event is bound to an exact location, such as a building, a street, or a county, and usually lasts for a short period." (p.2)
Q53	"The selected clusters cover events mainly from four crisis domains, including wildfire, local fire, storm, and traffic." (p.5)
Q122	"We build CrisisLTLSum by limiting the search to four crisis domains: wildfire, local fire, traffic, and storm." (p.12)
Q124	"These states are selected as they are some of the areas subject to the most events falling into our crisis domains of interest. California, Victoria, and New South Wales are mainly selected due to the abundance of wildfires in hot seasons. Texas and Florida have been selected as they are both subject to wildfires and storm events during different months. Massachusetts and Illinois are selected first because those areas are frequently subjected to bad weather, and second as they contain big cities subject to traffic and local fire events alongside New York." (p.13)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-19	https://www.frontiersin.org/journals/earth-science/articles/10.3389/feart.2022.827236/full
WEB-20	CENAPRED semáforo phase vocabulary is essential output-ontology content for volcanic crises (https://www.cenapred.unam.mx/reportesVolcanesMX/)
WEB-26	Popocatepetl is at Amarillo Fase 2 with ongoing daily exhalations (167 on April 28, 2026), indicating persistent rather than spike-event tweet streams (https://www.infobae.com/mexico/2026/04/29/popocatepetl-tuvo-167-emisiones-este-28-de-abril/)

Strengths

- Three of the four covered domains (local fire, traffic, storms) have at least nominal applicability to CDMX urban crisis coverage [Q53, Q122], so timeline/summarization patterns learned on those event types may partially transfer for those specific event types in urban contexts.

ID	Evidence
Q53	"The selected clusters cover events mainly from four crisis domains, including wildfire, local fire, storm, and traffic." (p.5)
Q122	"We build CrisisLTLSum by limiting the search to four crisis domains: wildfire, local fire, traffic, and storm." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the target region include earthquakes, volcanic activity (with semáforo phase vocabulary), tropical-system flooding, civil unrest/protests, organized-crime-related public-safety events, and CDMX transportation infrastructure failures, in addition to fire/traffic/storm [WEB-7, WEB-20, elicitation A1].

IO-2: Yes — earthquakes, volcanic activity, tropical-system flooding, and civil unrest are entirely absent from the benchmark's four-domain ontology [Q1, Q53, Q122]. The Popocatepetl semáforo vocabulary and SASMEX seed-tweet conventions have no analog in the benchmark [WEB-7, WEB-20].

IO-3: No category in the benchmark is wholly irrelevant to the target region — wildfires, local fires, traffic, and storms all have regional analogs [Q122] — but the 'storm' category as operationalized via US keyword lists [Q126, Q127] carries US-specific damage profiles that may not transfer.

IO-4: Construct underrepresentation is severe: four of the highest-priority crisis types in the deployment area (earthquake, volcanic, tropical flooding, civil unrest) are missing. The 'local' scope definition [Q12] also excludes persistent volcanic activity, which is structurally central to CDMX coverage [WEB-26].

ID	Evidence
Q1	"CrisisLTLSum contains 1,000 crisis event timelines across four domains: wildfires, local fires, traffic, and storms." (p.1)
Q12	"Moreover, we focus on the extraction of local crisis events. The term 'local' indicates that an event is bound to an exact location, such as a building, a street, or a county, and usually lasts for a short period." (p.2)
Q53	"The selected clusters cover events mainly from four crisis domains, including wildfire, local fire, storm, and traffic." (p.5)
Q122	"We build CrisisLTLSum by limiting the search to four crisis domains: wildfire, local fire, traffic, and storm." (p.12)
Q124	"These states are selected as they are some of the areas subject to the most events falling into our crisis domains of interest. California, Victoria, and New South Wales are mainly selected due to the abundance of wildfires in hot seasons. Texas and Florida have been selected as they are both subject to wildfires and storm events during different months. Massachusetts and Illinois are selected first because those areas are frequently subjected to bad weather, and second as they contain big cities subject to traffic and local fire events alongside New York." (p.13)
Q126	"We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past." (p.13)
Q127	"Please note that these lists are generic for each category and not specific to unique events. For instance, instead of defining keywords specific to an event called 'Hurricane Ida', our keywords list includes phrases such as 'fallen tree', 'building collapse', or 'storm'." (p.13)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-20	CENAPRED semáforo phase vocabulary is essential output-ontology content for volcanic crises (https://www.cenapred.unam.mx/reportesVolcanesMX/)
WEB-26	Popocatepetl is at Amarillo Fase 2 with ongoing daily exhalations (167 on April 28, 2026), indicating persistent rather than spike-event tweet streams (https://www.infobae.com/mexico/2026/04/29/popocatepetl-tuvo-167-emisiones-este-28-de-abril/)

Information Gaps

- Frequency of cartel/organized-crime crisis events in tweet streams is undocumented in indexed sources
- Relative weighting of missing categories vs. the four covered ones in actual journalist workflow not quantified

Requires Expert Verification

- Whether outlet-specific editorial priorities further alter category importance ranking

Remediation

Gap 1: Earthquake, volcanic activity, tropical-system flooding, civil unrest, and CDMX transportation infrastructure failures are absent from the four-domain taxonomy [Q122]

Recommendation: Extend the ontology with these crisis types and curate Spanish-language seed-tweet/keyword lists for each, including SASMEX (#TenemosSismo, #AlertaSismica) [WEB-7] and CENAPRED semáforo vocabulary [WEB-20]. Recognize persistent (e.g., volcanic) vs. spike-event temporal profiles [WEB-26] in the 'local' definition [Q12].

Gap 2: The 'local' scope definition (building/street/county, short period) [Q12] excludes persistent volcanic activity and multi-day tropical systems

Recommendation: Generalize the temporal-scope definition to support persistent and multi-phase events, with explicit handling of semáforo phase transitions and tropical-system alert lifecycles.

ID	Evidence
Q12	"Moreover, we focus on the extraction of local crisis events. The term "local" indicates that an event is bound to an exact location, such as a building, a street, or a county, and usually lasts for a short period." (p.2)
Q122	"We build CrisisLTLSum by limiting the search to four crisis domains: wildfire, local fire, traffic, and storm." (p.12)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-20	CENAPRED semáforo phase vocabulary is essential output-ontology content for volcanic crises (https://www.cenapred.unam.mx/reportesVolcanesMX/)
WEB-26	Popocatepetl is at Amarillo Fase 2 with ongoing daily exhalations (167 on April 28, 2026), indicating persistent rather than spike-event tweet streams (https://www.infobae.com/mexico/2026/04/29/popocatepetl-tuvo-167-emisiones-este-28-de-abril/)

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: CRISISLTLSUM | Context: Mexico City and Central America Bilingual Crisis Journalism

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

All input content was collected from seven US states and two Australian states during 2020–2021 [Q123], using English-language keyword lists curated from English news and social media [Q126, Q127], English Twitter NER models [Q35], and OpenStreetMap location augmentation anchored to US/Australian gazetteers [Q30, Q36]. The authors themselves note that replication for other languages or regions would be difficult [Q119]. For the target deployment, no Mexican or Central American content is present; Mexican administrative geography (16 alcaldías, colonias, 32 estados, municipios) and Central American departamentos/municipios are entirely absent. The web research confirms that no Spanish-language crisis tweet corpus anchored to Mexican or Central American events exists [WEB-3, WEB-4, WEB-6]. Mexican Spanish Twitter conventions (diacritic dropping, regional vocabulary like 'temblor'/'sismo'/'edomex', SASMEX hashtags #TenemosSismo, voseo in Central America, code-switching at the border) are entirely unrepresented in the benchmark's content [WEB-7, WEB-8]. The keyword-filtering pipeline is also acknowledged as not comprehensive [Q132], and its effectiveness depends on English-curated lists [Q128].

ID	Evidence
Q30	"The location filtering relies on a list of location candidates created by gathering cities, towns, and famous neighborhoods in a big area of interest. A tweet is considered relevant to our area of interest if 1) the text mentions one of the candidates, 2) the geo-tag matches the area of interest, or 3) the user location matches the area of interest." (p.3)
Q35	"Since location mentions are crucial in extracting local events and existing models have low performance detecting them from noisy tweets text, we further developed a BERT-based NER model tuned on Twitter data to detect location mentions." (p.4)
Q36	"Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses." (p.4)
Q119	"Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies." (p.10)
Q123	"We collect CrisisLTLSum across seven states in the USA and two states of Australia during 2020 and 2021. The selected states in the USA are New York, Illinois, Massachusetts, California, Texas, and Florida. From Australia, we have selected the states of New South Wales and Victoria." (p.12)
Q126	"We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past." (p.13)
Q127	"Please note that these lists are generic for each category and not specific to unique events. For instance, instead of defining keywords specific to an event called "Hurricane Ida", our keywords list includes phrases such as "fallen tree", "building collapse", or "storm"." (p.13)
Q128	"The quality of the keywords list is crucial to the final quality of the generated timelines, and we polish this list multiple times based on small sets of experiments before using them for the final task." (p.13)
Q132	"This approach will not be comprehensive or exhaustive but rather representative of each crisis domain. Improving this method to be more encompassing is an area for future research." (p.13)
WEB-3	HumAID is English-only despite including a 2017 Mexico earthquake subset — confirming the gap in Spanish-language Mexican crisis tweet content (https://crisisnlp.qcri.org/humaid_dataset.html)
WEB-4	Curiel et al. (PMC 2019) is the only known Mexican-Spanish crisis NER/geolocation resource and is not a timeline/summarization dataset (https://pmc.ncbi.nlm.nih.gov/articles/PMC6484392/)
WEB-6	https://github.com/wangcongcong123/crisis_nlp_progress/blob/master/datasets/TREC-IS.md
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)

WEB-8	@SASMEX X account provides structured earthquake seed tweets with no analog in benchmark content (https://x.com/sasmex)
-------	--

Strengths

- The general principle of geographic and temporal focus (specific area × domain × time period) [Q29, Q32] is methodologically transferable, even though the specific filters used are US-anchored.
- The benchmark explicitly acknowledges its non-replicability for other languages/regions [Q119], providing transparency that supports informed adaptation decisions.

ID	Evidence
Q29	"We limit the incoming tweets to specific geographical areas, periods, and domains of interest." (p.3)
Q32	"The combinations of (area a, domain d, time t) are manually selected so that the events of type d at location a are more frequent during time period t." (p.3)
Q119	"Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — target inputs require Mexican Spanish dialectal knowledge, alcaldía/colonia/municipio toponymy, Central American voseo, indigenous-language place names (Nahuatl, Zapotec, Maya, Mam, K'iche'), and SASMEX/CENAPRED hashtag conventions, none of which are present in benchmark content [Q123, WEB-4, WEB-7].

IC-2: No — the benchmark's English-language US/Australian content does not align culturally or linguistically with bilingual Mexican/Central American crisis journalism [Q123, Q126].

IC-3: Yes — input content is filtered through English-language keyword lists [Q126, Q127] and US/Australian location lists [Q30] that would not transfer to Spanish-language Mexican crisis tweets. The OpenStreetMap-based location augmentation [Q36] requires Spanish-language gazetteer adaptation not provided.

IC-4: INSUFFICIENT DOCUMENTATION — no Spanish-speaking annotators or regional reviewers were involved in content selection [Q52, Q61]. Recruitment of Mexican/Central American journalists or linguists would be required to identify culturally sensitive instances.

IC-5: Content validity is severely compromised: the benchmark contains zero Spanish-language content, zero Mexican/Central American geographic anchors, and uses an English-only NER + OSM pipeline that the authors explicitly note is not portable [Q119].

ID	Evidence
Q30	"The location filtering relies on a list of location candidates created by gathering cities, towns, and famous neighborhoods in a big area of interest. A tweet is considered relevant to our area of interest if 1) the text mentions one of the candidates, 2) the geo-tag matches the area of interest, or 3) the user location matches the area of interest." (p.3)
Q36	"Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses." (p.4)
Q52	"We, authors of this work, manually selected 1,000 clusters that contain enough tweets describing how a crisis event evolves, while specifying the "seed tweet" (i.e. the first observed post that describes the ongoing event) of each timeline." (p.5)

Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q119	"Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies." (p.10)
Q123	"We collect CrisisLTLSum across seven states in the USA and two states of Australia during 2020 and 2021. The selected states in the USA are New York, Illinois, Massachusetts, California, Texas, and Florida. From Australia, we have selected the states of New South Wales and Victoria." (p.12)
Q126	"We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past." (p.13)
Q127	"Please note that these lists are generic for each category and not specific to unique events. For instance, instead of defining keywords specific to an event called "Hurricane Ida", our keywords list includes phrases such as "fallen tree", "building collapse", or "storm". " (p.13)
Q132	"This approach will not be comprehensive or exhaustive but rather representative of each crisis domain. Improving this method to be more encompassing is an area for future research." (p.13)
WEB-3	HumAID is English-only despite including a 2017 Mexico earthquake subset — confirming the gap in Spanish-language Mexican crisis tweet content (https://crisisnlp.qcri.org/humaid_dataset.html)
WEB-4	Curiel et al. (PMC 2019) is the only known Mexican-Spanish crisis NER/geolocation resource and is not a timeline/summarization dataset (https://pmc.ncbi.nlm.nih.gov/articles/PMC6484392/)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)

Information Gaps

- Frequency of indigenous-language place names in crisis tweets unknown
- Voseo penetration in Central American crisis tweets unquantified

Requires Expert Verification

- Mexican/Central American journalist review of which colonia/municipio/landmark references are most frequent in crisis tweet streams
- Linguist review of code-switching patterns at US-Mexico border

Remediation

Gap 1: Zero Spanish-language Mexican/Central American content; English NER + English-OSM pipeline is non-portable [Q35, Q36, Q119]

Recommendation: Build or commission a Spanish-language crisis tweet corpus anchored to Mexican (CDMX alcaldías, key estados) and Central American (departamentos/municipios) events. Adapt the Curiel et al. (PMC 2019) Mexican-Spanish NER pipeline [WEB-4] for georesolution, and integrate a Spanish-language gazetteer that includes colonias, alcaldías, and indigenous-language place names.

ID	Evidence
Q35	"Since location mentions are crucial in extracting local events and existing models have low performance detecting them from noisy tweets text, we further developed a BERT-based NER model tuned on Twitter data to detect location mentions." (p.4)
Q36	"Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses." (p.4)
Q119	"Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies." (p.10)
WEB-4	Curiel et al. (PMC 2019) is the only known Mexican-Spanish crisis NER/geolocation resource and is not a timeline/summarization dataset (https://pmc.ncbi.nlm.nih.gov/articles/PMC6484392/)

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: CRISISLTLSUM | Context: Mexico City and Central America Bilingual Crisis Journalism

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The modality match is good: both the benchmark and target deployment operate on text-only tweets [Q2, Q18], and the elicitation explicitly marks this dimension as LOWER priority because no modality mismatch exists. However, the linguistic form of inputs is exclusively English [Q35, Q86], with preprocessing (lemmatization, lowercasing, fuzzy duplicate matching) implemented for English [Q129, Q136], and a 512-token BERT ceiling [Q92]. For a Spanish-language deployment, tokenization, lemmatization, and morphological handling must accommodate Spanish diacritics, accented-character variants, and informal diacritic-dropping in tweets — none of which is documented as supported. The 512-token sequence ceiling and tweet-text format are technically appropriate for the target. Per the elicitation, the linguistic-form mismatch is captured under IC and OF rather than IF; the modality dimension itself is well-aligned.

ID	Evidence
Q2	"We built CrisisLTLSum using a semi-automated cluster-then-refine approach to collect data from the public Twitter stream." (p.1)
Q18	"CrisisLTLSum is developed through a two-step semi-automated process to create 1,000 local crisis timelines from the public Twitter stream." (p.2)
Q35	"Since location mentions are crucial in extracting local events and existing models have low performance detecting them from noisy tweets text, we further developed a BERT-based NER model tuned on Twitter data to detect location mentions." (p.4)
Q86	"For our first sentence-level classification model, we fine-tune BERT (Devlin et al., 2019) on the tweet sequences where every training example would have the form of "t0 [SEP] si"; where si ∈ S." (p.7)
Q92	"Although the BERT-based sequence classification model has a limitation when extracting long timelines due to its limited size of 512 positional embeddings, it performs better than the BERT-GRU sequence labeling model." (p.7)
Q129	"To avoid making a long list of keywords and ensure that different lexical formats of the same word are still considered in the keyword matching process, we maintain both a lower-cased and a lemmatized version of the keywords and the tweet's text." (p.13)
Q136	"Here, the goal is to remove tweets from the same cluster with identical or similar text. The duplicate removal relies on a fuzzy string sequence-matching technique to compute the similarity between a pair of tweets. We simply go over each tweet sorted in chronological order and remove the ones that match any of the previous tweets in the same cluster with a matching score higher than dmatch." (p.13)

Strengths

- Text-only tweet modality matches the target deployment exactly [Q2, Q18, elicitation IF-priority=LOWER].
- Standard tweet preprocessing pipeline (lemmatization, deduplication) [Q129, Q136] is methodologically applicable, even if its specific implementation is English-only.
- BERT-style 512-token sequence handling [Q86, Q92] is compatible with multilingual transformer backbones used for Spanish.

ID	Evidence
Q2	"We built CrisisLTLSum using a semi-automated cluster-then-refine approach to collect data from the public Twitter stream." (p.1)
Q18	"CrisisLTLSum is developed through a two-step semi-automated process to create 1,000 local crisis timelines from the public Twitter stream." (p.2)
Q86	"For our first sentence-level classification model, we fine-tune BERT (Devlin et al., 2019) on the tweet sequences where every training example would have the form of "t0 [SEP] si"; where si ∈ S." (p.7)

Q92	"Although the BERT-based sequence classification model has a limitation when extracting long timelines due to its limited size of 512 positional embeddings, it performs better than the BERT-GRU sequence labeling model." (p.7)
Q129	"To avoid making a long list of keywords and ensure that different lexical formats of the same word are still considered in the keyword matching process, we maintain both a lower-cased and a lemittized version of the keywords and the tweet's text." (p.13)
Q136	"Here, the goal is to remove tweets from the same cluster with identical or similar text. The duplicate removal relies on a fuzzy string sequence-matching technique to compute the similarity between a pair of tweets. We simply go over each tweet sorted in chronological order and remove the ones that match any of the previous tweets in the same cluster with a matching score higher than dmatch." (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution match is acceptable at the modality level (text tweets) [Q2, Q18]. Within-modality differences (English vs. Spanish character set, diacritic handling, tweet-register orthography) require Spanish-specific preprocessing not documented in the benchmark [Q129].

IF-2: Yes — Mexican mobile/Twitter infrastructure supports the same text-tweet capture format. Mexico has ~152M mobile connections (1.2 phones/person) [WEB-14], indicating high source-tweet generation capacity. Central America country-level data was not retrieved [WEB-15].

IF-3: Domain-specific differences include: SASMEX-formatted seed tweets have a structured machine-readable form [WEB-7, WEB-8] that the benchmark's pipeline does not specifically handle; Spanish accented-character handling is required but undocumented; tweet-register diacritic-dropping is a known phenomenon requiring graceful handling.

IF-4: Form mismatches at the modality level are minor. The substantive linguistic-form mismatches (English-only preprocessing, English NER) materially affect IC and OF rather than IF per the elicitation framing.

ID	Evidence
Q2	"We built CrisisLTLSum using a semi-automated cluster-then-refine approach to collect data from the public Twitter stream." (p.1)
Q18	"CrisisLTLSum is developed through a two-step semi-automated process to create 1,000 local crisis timelines from the public Twitter stream." (p.2)
Q86	"For our first sentence-level classification model, we fine-tune BERT (Devlin et al., 2019) on the tweet sequences where every training example would have the form of "t0 [SEP] si"; where si ∈ S." (p.7)
Q92	"Although the BERT-based sequence classification model has a limitation when extracting long timelines due to its limited size of 512 positional embeddings, it performs better than the BERT-GRU sequence labeling model." (p.7)
Q129	"To avoid making a long list of keywords and ensure that different lexical formats of the same word are still considered in the keyword matching process, we maintain both a lower-cased and a lemittized version of the keywords and the tweet's text." (p.13)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-8	@SASMEX X account provides structured earthquake seed tweets with no analog in benchmark content (https://x.com/sasmex)
WEB-14	Mexico mobile-phone penetration ~1.2 phones/person supports tweet-stream input form expectations (https://www.worlddata.info/america/mexico/telecommunication.php)
WEB-15	https://datahub.itu.int/data/?e=1&i=11624

Information Gaps

- Country-specific Central America mobile/internet penetration figures not retrieved
- Spanish tweet diacritic-dropping rates in crisis contexts not quantified

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: CRISISLTLSUM | Context: Mexico City and Central America Bilingual Crisis Journalism

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output ontology consists of (a) a binary timeline-membership label governed by three conditions — relevant, informative, non-repetitive [Q58] — and (b) a four-axis summary quality rubric (Coherence, Accuracy, Coverage, Overall Quality) rated 1–5 [Q106, Q151]. The three relevance conditions are conceptually generalizable, but were validated only on US English tweets by Anglophone annotators [Q61], and cross-lingual transfer validity to Mexican Spanish tweets covering different event types is unestablished. Critically, the Coherence rubric is explicitly anchored to English: 'A summary is coherent if when read by itself, it's easy to understand and free of English errors' [Q153, Q154]. This makes the coherence axis structurally invalid for Spanish-language output. Additionally, the informativeness criterion encodes US-centric judgments about what constitutes a meaningful crisis update; the Mexican context includes documented genres (community solidarity posts, citizen verification posts like #Verificado19S) that have distinct informativeness profiles [WEB-23, WEB-24]. No category captures editorial dimensions (source attribution to CENAPRED/SSN/SASMEX, casualty framing, AP Español alignment) that are central to the target deployment.

ID	Evidence
Q58	"A tweet is labeled as part of the timeline only if it satisfies all the following three conditions: relevant: talks about the same event indicated in the seed tweet, informative: provides facts about the event but not only contains personal points of view, not repetitive: brings in new information." (p.5)
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)
Q151	"During this task, you will be reading a timeline that consists of a sequence of tweets describing an event and eight different summaries for the event in the timeline. You will rate the quality of each of the eight summaries by four axes: coherence, accuracy, coverage, and overall quality." (p.21)
Q153	"For each summary, answer the question 'how coherent is the summary on its own?' on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)
Q154	"Score of 1: The summary is impossible to understand. Score of 2: The summary has many mistakes or confusing phrasing. Score of 3: The summary has some mistakes or confusing phrasing that made it hard to understand. Score of 4: The summary is mostly clear with only a few minor mistakes or confusing phrasing. Score of 5: The summary is perfectly clear." (p.21)
WEB-23	#Verificado19S citizen-verification genre would be judged differently by Mexican journalists than US MTurk workers (https://arxiv.org/abs/2007.05848)
WEB-24	https://link.springer.com/article/10.1007/s00779-020-01411-5

Strengths

- The three-criterion relevance/informativeness/non-redundancy schema [Q58] is conceptually portable and provides a transferable starting point for Spanish-language label adaptation.
- The four-axis summary rubric (Coherence, Accuracy, Coverage, Overall) [Q106] decomposes summary quality in a way that supports targeted adaptation — Accuracy and Coverage axes [Q155, Q157] are linguistically more neutral than

Coherence.

ID	Evidence
Q58	"A tweet is labeled as part of the timeline only if it satisfies all the following three conditions: relevant: talks about the same event indicated in the seed tweet, informative: provides facts about the event but not only contains personal points of view, not repetitive: brings in new information." (p.5)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)
Q155	"For each summary, answer the question "Does the factual information in the summary accurately match the information in the timeline?" on a scale of 1 to 5. A summary is accurate if it doesn't say things that aren't in the timeline, it doesn't mix up information, and generally is not misleading." (p.21)
Q157	"For each summary, answer the question "How well does the summary cover the important information in the timeline?" on a scale of 1 to 5. A good summary has good coverage if it mentions the main information from the timeline that's important to understand the event. A summary has poor coverage if someone reading only the summary would be missing several important pieces of information about the event in the timeline." (p.21)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The binary timeline-membership label and four-axis quality rubric are partially relevant, but the Coherence axis is explicitly English-anchored [Q153] and the relevance/informativeness conditions [Q58] were validated only on US tweets.

OO-2: Missing categories include: source-attribution quality (whether CENAPRED/SSN/CNPC are cited correctly), AP Español register conformance, verification-tweet detection (a specific genre in #19S contexts) [WEB-23], and seismic/volcanic alert-vocabulary correctness (semáforo phases, SASMEX hashtag usage) [WEB-7, WEB-20].

OO-3: The Coherence axis encodes 'free of English errors' [Q153] as the criterion, which directly encodes Anglophone language norms. The informativeness criterion [Q58] implicitly encodes US news-reading conventions about what counts as a 'fact' versus 'personal point of view,' and was validated only by Anglophone annotators [Q61].

OO-4: Significant misalignment exists. Stakeholder-driven taxonomy redesign is warranted: a Spanish-language coherence rubric, a register/style axis aligned with AP Español, and addition of source-attribution and verification-correctness axes are needed.

OO-5: Structural validity is compromised by the English-anchored coherence rubric; content validity is compromised by missing axes for register, attribution, and verification quality. External validity transfer is therefore not supported.

ID	Evidence
Q58	"A tweet is labeled as part of the timeline only if it satisfies all the following three conditions: relevant: talks about the same event indicated in the seed tweet, informative: provides facts about the event but not only contains personal points of view, not repetitive: brings in new information." (p.5)
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)

Q151	"During this task, you will be reading a timeline that consists of a sequence of tweets describing an event and eight different summaries for the event in the timeline. You will rate the quality of each of the eight summaries by four axes: coherence, accuracy, coverage, and overall quality." (p.21)
Q153	"For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)
Q154	"Score of 1: The summary is impossible to understand. Score of 2: The summary has many mistakes or confusing phrasing. Score of 3: The summary has some mistakes or confusing phrasing that made it hard to understand. Score of 4: The summary is mostly clear with only a few minor mistakes or confusing phrasing. Score of 5: The summary is perfectly clear." (p.21)
Q155	"For each summary, answer the question "Does the factual information in the summary accurately match the information in the timeline?" on a scale of 1 to 5. A summary is accurate if it doesn't say things that aren't in the timeline, it doesn't mix up information, and generally is not misleading." (p.21)
Q157	"For each summary, answer the question "How well does the summary cover the important information in the timeline?" on a scale of 1 to 5. A good summary has good coverage if it mentions the main information from the timeline that's important to understand the event. A summary has poor coverage if someone reading only the summary would be missing several important pieces of information about the event in the timeline." (p.21)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-20	CENAPRED semáforo phase vocabulary is essential output-ontology content for volcanic crises (https://www.cenapred.unam.mx/reportesVolcanesMX/)
WEB-23	#Verificado19S citizen-verification genre would be judged differently by Mexican journalists than US MTurk workers (https://arxiv.org/abs/2007.05848)

Information Gaps

- AP Español operationalizable criteria not openly accessible — flagged for expert elicitation
- Whether Mexican journalists' relevance judgments diverge systematically from the benchmark's three-criterion definition is empirically unverified

Requires Expert Verification

- Mexican journalist review of relevance/informativeness boundaries on Spanish crisis tweets
- AP Español editor input on register and source-attribution rubric

Remediation

Gap 1: Coherence rubric is explicitly anchored to English [Q153]; missing axes for source attribution, AP Español register, and verification-tweet detection

Recommendation: Construct a Spanish coherence rubric referencing Spanish-language writing norms; add an AP Español-aligned register axis (requires expert elicitation from AP Español editors) and a source-attribution axis covering CENAPRED/SSN/CNPC/SASMEX [WEB-1, WEB-7]; add a verification-quality axis informed by the #Verificado19S model [WEB-23].

ID	Evidence
Q153	"For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)
WEB-1	CENAPRED is the authoritative Mexican source for crisis attribution — distinct from US norms (https://en.wikipedia.org/wiki/CENAPRED)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-23	#Verificado19S citizen-verification genre would be judged differently by Mexican journalists than US MTurk workers (https://arxiv.org/abs/2007.05848)

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: CRISISLTLSUM | Context: Mexico City and Central America Bilingual Crisis Journalism

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Ground-truth labels were determined by majority vote among three MTurk workers **[Q70]** recruited under QC1, which restricted workers to 'countries where native English speakers are most likely to be found' **[Q61, Q105]**. Reference summaries were written by the two workers who agreed most on extraction labels **[Q87, Q88]**. While inter-annotator agreement was high (90.06% **[Q77]**, 91.77% against experts **[Q78, Q79]**), no Spanish-speaking annotators, no journalists, and no individuals with Latin American crisis knowledge contributed to ground truth. The elicitation explicitly confirms that journalistic relevance judgments, attribution norms, and summary scope differ from US conventions (Q3/A3). Therefore the labels and reference summaries cannot be assumed to reflect Mexican/Central American journalistic stakeholder perspectives. The aggregation method (majority vote on binary labels **[Q70]**; selecting the two most-agreeing workers' summaries **[Q87, Q88]**) further erases minority perspectives — a particular concern when no annotator represents the target population at all. Documented Mexican-context phenomena like #Verificado19S verification posts and community solidarity posts **[WEB-23]** would likely be judged differently by Mexican journalists than by Anglophone MTurk workers.

ID	Evidence
Q3	"A timeline is a chronologically sorted set of posts, where each brings in new information or updates about an ongoing event (such as a fire, storm, or traffic incident)." (p.1)
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q70	"To make the ground-truth on whether each tweet is part of the timeline or not, we take the majority label among the three workers." (p.6)
Q77	"The average timeline-level agreement between those two workers is 90.06%." (p.6)
Q78	"We also explore a more profound analysis by comparing the workers' annotations on 20% of the timelines against the annotations of experts." (p.6)
Q79	"The average timeline-level agreement rate of this analysis is 91.77%." (p.6)
Q87	"Since each timeline in our dataset is annotated by three workers (§3.2), it includes three summaries." (p.7)
Q88	"To adapt the timeline summarization task to this setting, we pick the two summaries written by the two workers who agree the most based on the timeline extraction labels they assign to the tweets in each timeline." (p.7)
Q105	"Following a similar process in 3.2, we recruit a group of workers from MTurk with the same Location restriction (QC1), who can pass our pilot study as a qualification test." (p.8)
WEB-23	#Verificado19S citizen-verification genre would be judged differently by Mexican journalists than US MTurk workers (https://arxiv.org/abs/2007.05848)

Strengths

- Documentation of annotator quality control (QC1–QC4) is detailed **[Q60–Q66]**, providing transparency about who did label and enabling targeted re-annotation planning.
- High inter-annotator agreement (90.06% **[Q77]**) and expert validation (91.77% **[Q79]**) demonstrate internal consistency of labels within the Anglophone population.
- Compensation transparency (\$3/HIT, >\$10/hour effective rate) **[Q67, Q68]** is documented.

ID	Evidence
----	----------

Q60	"Following prior quality control practices (Briakou et al., 2021), we use multiple quality control (QC) steps to ensure the recruitment of high-quality annotators." (p.5)
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q62	"Next, we recruit annotators who pass our qualification test (QC2), where we ask them to annotate 3 timelines." (p.5)
Q63	"We run several small pilot tasks, each with a replication factor of nine." (p.5)
Q64	"We check annotators' performance on timeline extraction task against experts' labels and have experts manually review (QC3) annotators' summary quality." (p.5)
Q65	"Only workers passing all the quality control steps contribute to the final task." (p.5)
Q66	"During the final task, we perform regular quality checks (QC4), and only use workers who consistently perform well." (p.5)
Q67	"We compensate the workers at a rate of \$3 per HIT for the task." (p.5)
Q68	"Each batch of tasks is followed by a one-time bonus that makes the final rate over \$10 per hour." (p.5)
Q77	"The average timeline-level agreement between those two workers is 90.06%." (p.6)
Q79	"The average timeline-level agreement rate of this analysis is 91.77%." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — ground-truth labels reflect Anglophone MTurk worker judgments [Q61, Q70], not Mexican/Central American journalist perspectives. The elicitation (A3) confirms divergent relevance and attribution norms.

OC-2: Substantial disagreement is likely. Mexican journalists operating under AP Español norms apply distinct attribution standards (CENAPRED/SSN/CNPC) [WEB-1, WEB-12] and may judge community solidarity posts and verification posts [WEB-23] as more informative than US MTurk workers would.

OC-3: Annotator demographics are documented only at the level of 'countries where native English speakers are most likely to be found' [Q61, Q105]. No information on age, gender, occupation, or journalistic background is provided. No Datasheet or Data Statement is referenced.

OC-4: Re-annotation by a representative pool — Spanish-speaking journalists familiar with Mexican/Central American crisis contexts — is required for validity in the target deployment [elicitation A3].

OC-5: Aggregation by majority vote on binary labels [Q70] and selection of the two most-agreeing workers' summaries [Q87, Q88] erases minority perspectives. When the entire annotator pool is non-representative of the target population, this aggregation compounds rather than mitigates the validity gap.

OC-6: Convergent validity with regional stakeholders is unsupported; external validity to Mexican journalist judgments cannot be assumed.

ID	Evidence
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)

Q70	"To make the ground-truth on whether each tweet is part of the timeline or not, we take the majority label among the three workers." (p.6)
Q87	"Since each timeline in our dataset is annotated by three workers (§3.2), it includes three summaries." (p.7)
Q88	"To adapt the timeline summarization task to this setting, we pick the two summaries written by the two workers who agree the most based on the timeline extraction labels they assign to the tweets in each timeline." (p.7)
Q105	"Following a similar process in 3.2, we recruit a group of workers from MTurk with the same Location restriction (QC1), who can pass our pilot study as a qualification test." (p.8)
WEB-1	CENAPRED is the authoritative Mexican source for crisis attribution — distinct from US norms (https://en.wikipedia.org/wiki/CENAPRED)
WEB-12	Central American national civil-protection agencies (CONRED, COPECO, DGPC, SINAPRED, CNE, SINAPROC) define the local source-attribution ecosystem (https://whndn.org/portfolio-item/disaster-response-services-tools/)
WEB-23	#Verificado19S citizen-verification genre would be judged differently by Mexican journalists than US MTurk workers (https://arxiv.org/abs/2007.05848)

Information Gaps

- Specific MTurk worker demographics beyond country of residence are undocumented
- The magnitude of disagreement between Anglophone MTurk workers and Mexican journalists is empirically unmeasured

Requires Expert Verification

- Pilot re-annotation study with Mexican journalists to quantify disagreement rates on representative Spanish-language crisis tweets
- AP Español editor review of summary references for register and attribution conformance

Remediation

Gap 1: Reference labels and summaries authored entirely by Anglophone MTurk workers under QC1 location restriction [Q61]; no journalist or Mexican stakeholder representation

Recommendation: Re-annotate a representative subset using Mexican Spanish-speaking journalists familiar with Mexican/Central American crisis contexts and AP Español. Quantify disagreement against the original labels and report convergent-validity statistics. Avoid majority-vote aggregation that erases minority perspectives when annotator pools are small.

ID	Evidence
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)

Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: CRISISLTLSUM | Context: Mexico City and Central America Bilingual Crisis Journalism

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark evaluates only English-language output: timeline-extraction accuracy is measured against expert annotations on English tweets [Q91, Q94, Q96]; timeline summaries are evaluated using SacreROUGE [Q98, Q100] against English-language reference summaries written by Anglophone MTurk workers [Q99]; the human-evaluation rubric is written in English and scores coherence against English-language writing norms [Q106, Q153]. The target deployment requires bilingual output (Mexican Spanish + American English) with the elicitation specifying that English summaries should be more directly publication-ready and Spanish summaries adapted by journalists (A4). The benchmark provides no Spanish-language reference summaries, no Spanish coherence rubric, no bilingual evaluation protocol, no AP Español alignment metric, and no register-appropriateness scoring. The web research further documents that ROUGE correlates poorly with human judgments for Spanish summarization, with BERTScore-precision (multilingual) and LLM-as-judge being current best practice [WEB-17, WEB-18] — none of which the benchmark uses. The mismatch with the deployment's bilingual output requirement is total for Spanish.

ID	Evidence
Q91	"For evaluation, we use the average timeline-level accuracy." (p.7)
Q94	"For timeline extraction results on the TEST set, we compare both the human-level performance and the best model's performance against experts' annotations." (p.8)
Q96	"The best timeline extraction model achieves 73.51 average timeline accuracy, whereas the human-level performance is 88.98." (p.8)
Q98	"We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004)." (p.8)
Q99	"The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels." (p.8)
Q100	"We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores." (p.8)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)
Q153	"For each summary, answer the question 'how coherent is the summary on its own?' on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)
WEB-17	Standard ROUGE variants correlate weakly with human judgments for Spanish summarization; BERTScore-precision with multilingual models is recommended (https://arxiv.org/pdf/2503.17039)
WEB-18	LLM-as-judge approaches show strong correlation with human judgments in multilingual summarization evaluation (https://aclanthology.org/2024.emnlp-main.1085.pdf)

Strengths

- The English-language output evaluation [Q98, Q106] is methodologically sound and directly applicable to the deployment's English-output evaluation needs, where English summaries are expected to be higher-quality and more publication-ready (elicitation A4).
- Multi-reference ROUGE [Q99, Q100] and the four-axis human evaluation [Q106] together provide a defensible English evaluation framework that can be reused for the English half of bilingual output.

ID	Evidence
Q98	"We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004)." (p.8)
Q99	"The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels." (p.8)
Q100	"We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores." (p.8)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Partial match — output modality (free-form text summary) matches deployment requirements, but the language coverage (English-only) misses 50% of the bilingual requirement [[Q99](#), [Q106](#); [elicitation A4](#)].

OF-2: Not applicable — deployment does not require text-to-speech output (journalists work with text artifacts).

OF-3: Not applicable — target users are professional journalists with high literacy; no accessibility constraints alter the form requirement.

OF-4: Form mismatch: complete absence of Spanish-language summarization evaluation [[Q99](#), [Q106](#), [Q153](#)] directly contradicts the bilingual deployment requirement, severely harming external validity for the Spanish output channel.

ID	Evidence
Q98	"We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004)." (p.8)
Q99	"The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels." (p.8)
Q100	"We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores." (p.8)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)
Q153	"For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)
WEB-17	Standard ROUGE variants correlate weakly with human judgments for Spanish summarization; BERTScore-precision with multilingual models is recommended (https://arxiv.org/pdf/2503.17039)
WEB-18	LLM-as-judge approaches show strong correlation with human judgments in multilingual summarization evaluation (https://aclanthology.org/2024.emnlp-main.1085.pdf)

Information Gaps

- Operationalizable AP Español rubric criteria not openly accessible
- No Spanish crisis-summarization reference dataset exists to enable bilingual evaluation construction

Requires Expert Verification

- AP Español editor input to construct a Spanish coherence/register rubric
- Mexican journalist evaluation panel to validate any Spanish-language summarization quality framework

Remediation

Gap 1: No Spanish-language summarization evaluation; English-only ROUGE [Q99, Q100] and English coherence rubric [Q153]

Recommendation: Implement a bilingual evaluation protocol: BERTScore with multilingual backbone (XLM-R or mBERT) and LLM-as-judge for Spanish summaries [WEB-17, WEB-18], alongside human evaluation by Spanish-speaking journalists. Maintain the existing English ROUGE pipeline for the English output channel where it remains defensible.

ID	Evidence
Q99	“The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels.” (p.8)
Q100	“We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores.” (p.8)
Q153	“For each summary, answer the question “how coherent is the summary on its own?” on a scale 1 to 5. A summary is coherent if when read by itself, it’s easy to understand and free of English errors. A summary is not coherent if it’s difficult to understand what the summary is trying to say. Generally, it’s more important that the summary is understandable than it being free of grammar errors.” (p.21)
WEB-17	Standard ROUGE variants correlate weakly with human judgments for Spanish summarization; BERTScore-precision with multilingual models is recommended (https://arxiv.org/pdf/2503.17039)
WEB-18	LLM-as-judge approaches show strong correlation with human judgments in multilingual summarization evaluation (https://aclanthology.org/2024.emnlp-main.1085.pdf)